

The Marine Environment

The Marine Environment and Man-made Pollution

Exercise 1

When it comes to **mixing** oil and water, oceans suffer from far more than the occasional **devastating** oil spill.

An average oil change can **contaminate** a million gallons of fresh water.

Every year oily road runoff from a city of 5 million could **contain** as much oil as one large tanker spill.

Every year, bilge **cleaning** and other ship operations release millions of gallons of oil into navigable waters.

Air pollution can **place** hundreds of tons of hydrocarbons into the oceans each year.

Particles settle, and rain **washes** hydrocarbons from the air into the oceans.

Seepage from the ocean bottom and erosion of sedimentary rocks **releases** oil.

One big spill can **disrupt** sea and shore life for miles.

Offshore oil production can **cause** ocean oil pollution from spills and operational discharges.

Exercise 2

<Large> spills can be devastating.

Spills, even though a relatively <minor> case of ocean oil pollution.

Coral reefs and mangroves are <more> sensitive to oil than sandy beaches.

Intertidal zones are <the most> sensitive.

Oil exposure can cause <long-term> harm.

<Bottom-dwelling> fish are exposed to compounds released after oil spills.

Compounds are released <after> oil spills.

The 1986 spill was one of the <largest> in tropical North America.

Five years <later>, mangrove sediments still held oil.

It was thought that the mangroves would take fifty years to recover <fully>.

Exercise 3

More than half of all Americans <change> their own oil.

Collection centres and service stations <are accepting> used oil for recycling.

<To find out> about recycling oil....

Only about one-third of the used oil <is collected> and recycled.

MARPOL <is> shorthand for a United Nations treaty.

People <can contact> their local recycling authority.

Treaty <concerning> pollution problems

Oil collection and recycling programs <are now increasing> awareness of the hazards of dumping used oil.

More do-it-yourselfers <are doing> it right.

International cooperation <has greatly reduced> accidental and operational oil discharges.

MARPOL <was largely credited> for reducing oil pollution during the 1980s.

MARPOL <requires> installation and use of oil-pollution prevention equipment.

MARPOL <became> effective in 1983.

MARPOL <prohibits> discharges within certain distances of land.

Exercise 4

Toxic materials in sea-floor sediments become a hazard to marine life resulting in	a threat to organisms that live in and feed on bottom muds.
Long-lasting chemicals enter the food web giving rise to	contamination of fish and shellfish.
Industrial, agricultural, and household products end up in water leading to	toxic waste in waterways and coastal areas.
Digging out rivers, canals, and harbours causes	toxic materials to be dredged up leading to disposal problems.
Increasing ship sizes lead to	a continuing and growing demand for dredging.
Dredged material may contain unacceptable levels of contaminants and consequently	disposing of it safely is a serious problem.

Exercise 5

Factories, military bases, ships and sewage-treatment plants	contribute to toxic wastes in Puget Sound, Washington.
The Puget Sound Water Quality Authority has been overseeing efforts to	clean up contaminated sediments and control the sources of pollution.
State sediment-management standards and federal law	regulate methods for dredging, disposing, and cleanup of contaminated sediments.
TBT, or tributyl tin, is added to boat paints to	kill or repel barnacles that foul ships' hulls.

TBT, which seeps out of the paint into the water,	affects wildlife.
It is proven that the banning of TBT	encourages marine populations to return to the area.
Almost all the surviving animals	did not reproduce and had malformed shells.
After regulations reduced the use of TBT, oyster production	increased and shell deformation decreased dramatically.

Exercise 6

When <odds and ends> of life on land - particularly plastics - end up in the sea, they pose <hazards> to marine life.

Animals drown or strangle from becoming <tangled> in discarded or lost fishing gear.

Needles and <syringes>, washed from city streets and sewer systems after heavy rain storms, can wash up on beaches.

Estimated costs from lost tourism and <recreation> can run into billions.

Biologists who performed an autopsy on an <emaciated> male sperm whale <beached> at Sea Side Heights, New Jersey, found a party balloon.

Even desert isles become <dumps>.

Ducie Atoll is one of the most <remote> islands in the world.

An important breeding area for seabirds, the Ducie Atoll is <littered> with trash.

Exercise 7

International Coastal Cleanup, for example, was a statewide beach cleanup in Texas **in 1986**.

To handle mountains of plastic trash until ships reach port, the U.S. Navy has **recently** developed onboard processors to compact and sanitize plastic trash.

This equipment has been on all Navy ships **since 1999**, and has military and commercial applications worldwide.

When compressed into blocks, **thirty days'** worth can be stored in the space that would be filled by **one day's** worth of uncompacted plastic trash.

A **300-person** Navy destroyer generates **about 35 cubic feet** of plastic waste **per day**, which is compressed **into 5 to 6 disks**.

The Marine Environment

Hazardous and Noxious Substances

Exercise 1

A hazardous and noxious substance is defined as 'any substance other than oil which, if introduced into the marine environment is likely to create **hazards** to human health, to harm living **resources** and marine life, to damage **amenities** or to interfere with other **legitimate** uses of the sea'.

The OPRC-HNS Protocol **aims to** provide a global **framework** for international operation in **combating** major incidents or threats of marine pollution.

The OPRC-HNS Protocol ensures that ships carrying hazardous and noxious liquid substances are **covered** by preparedness and **response regimes** similar to those for oil incidents.

Ships **flying** the flag of a contracting state ('party') of the OPRC-HNS Protocol are required to carry a **pollution emergency plan** to deal specifically with incidents involving hazardous and noxious substances.

These substances can – when **spilled** – have a **harmful** impact on the marine environment and on marine **ecosystems**.

States which are **party** to the OPRC-HNS Protocol are required to develop a system for responding to HNS, including a designated **national contingency** plan.

IMO has developed a wide series of **tools** for dealing with incidents **involving** hazardous and noxious substances and meeting their obligations under the Protocol.

The OPRC-HNS Protocol also **contributes** to the increased protection of a state's **coastal zone** and marine environment, including **human health** and resources.

Exercise 2

Chemical **litter** and contamination can be traced from the **tropics** to the poles and from coastlines to **abyssal** depths.

Many chemicals have **physical properties** which complicate **containment** and management efforts in the case of incidents.

Some chemicals will **dissolve** in water, while highly **volatile** chemicals will **evaporate** rapidly over a limited area.

Highly volatile chemicals may produce a fire hazard extending beyond the area **covered by** the **pool** of spilled material.

Dangerous conditions may **prevail** downwind from such a **spill** over a considerable distance.

These different physical characteristics increase the risk of high costs **arising from** the environmental impacts of the spill, such as loss of **amenities**, closure of **fisheries** and tourism activities.

Some of these chemical substances are **carcinogenic** or highly **flammable**.

Exposure to toxic substances is a hazard to personnel conducting **salvage** and **response** operations for incidents involving hazardous and noxious substances.

The Marine Environment

Anti-fouling

Exercise 1

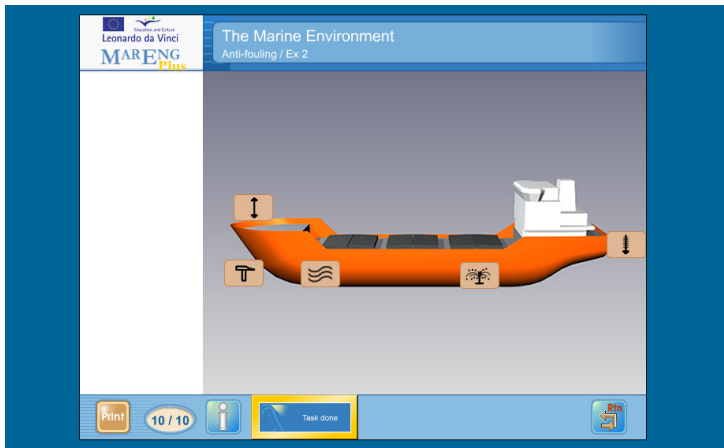
Marine vessels, buoys and **offshore** structures spend a large part of their working life partly **submerged** in water. As a result they become easily colonised by the many **micro-organisms** that live in the water, especially when **stationary**.

The extent of this colonisation, also known as **fouling**, depends on **salinity**, water temperature and productivity of the organic matter on which the organisms feed.

The marine organism can either be of plant origin such as weeds and **algae**, or of **animal** origin such as tubeworms and barnacles.

To protect the boat from colonisation, **anti-fouling** treatments such as paint are applied to the outer parts of the ship at **regular** intervals. The **coating** is designed to gradually leach biocide or a **toxin** into the marine environment. This slow **poisoning** prevents organisms from sticking to the hull.

Exercise 2



- A double arrow - symbol
- A hammer - symbol
- A wave - symbol
- A fountain - symbol
- A vertical arrow - symbol

Exercise 3

Copper results in reduced plant growth and **<a reduction>** in photosynthesis.

Teflon creates less **<friction>** on the surface.

The smoothness of Teflon reduces the **<settlement>** of organisms.

Teflon is used on racing **<yachts>**.

Tributyltin (TBT) lasts **<a lot longer than>** copper-based anti-foulings.

Teflon is not such an effective antifouling **<coating>**.

Silicon is a difficult material to work with because it is **<soft>**.

Copper is **<less effective than>** TBT.

Too high concentrations of copper prove to be extremely **<toxic>** to marine life.

Certain species of fish and other marine organisms such as marine invertebrates are **<sensitive>** to quite low levels of copper.

Copper is an element which **<naturally>** occurs.

The biggest copper polluters are industrial wastewater discharges and atmospheric **<deposition>**.

Organic biocides are less effective

The biggest copper polluters are industrial wastewater discharges

Tributyltin (TBT) can also be used on aluminium

Tributyltin (TBT) was banned by the IMO in 1987 because of its damage to shellfish

Tributyltin (TBT) used to be the most popular biocide

Organic

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Shipboard Marine Pollution Emergency Plans

Exercise 1

The MARPOL Convention requires that every oil tanker of 150 tons gross tonnage and above	carries on board a shipboard oil pollution emergency plan.
A shipboard marine pollution emergency plan for noxious liquid substances	should be combined with a shipboard oil pollution emergency plan.
One combined plan on board is more practical than	two separate plans in the event of an emergency.
The title of such a combined plan should be 'Shipboard marine pollution emergency plan'	in order to distinguish it from the two other emergency plans mentioned above.
Regulations of Annex I and Annex II of MARPOL	prescribe that the plans have to be in accordance with the guidelines of IMO.

It is relevant to note that shipboard oil pollution emergency plans which have already been approved	need not be modified.
These guidelines contain information for	the development of shipboard oil pollution emergency plans.
The main objectives of these guidelines are to assist shipowners	in preparing shipboard marine pollution emergency plans.
The main objectives of these guidelines are to assist governments	in enacting domestic laws.
In the interest of uniformity, governments	are requested to refer to these guidelines when preparing appropriate national regulations.

Exercise 2

The MARPOL **Convention** requires that every oil tanker of 150 tons **gross tonnage** and above carries on board a shipboard oil pollution **emergency plan approved by** the Flag State Administration.

New MARPOL Regulation requires that every ship of 150 tons gross tonnage and above **certified** to carry noxious liquid substances **in bulk** carries on board a shipboard marine pollution emergency plan for **noxious liquid** substances approved by the Administration.

One **combined** plan on board is more practical than two separate plans **in the event of** an emergency.

In this case the title of such a combined plan should be 'Shipboard marine pollution emergency plan' **in order to** distinguish it from the two other emergency plans **mentioned** above.

MARPOL regulations **prescribe** that the plans have to be **in accordance with** guidelines developed by the IMO.

It is **relevant** to note that shipboard oil pollution emergency plans which have already been approved by the Administration **need not** be modified.

The main **objectives** of these guidelines are to assist shipowners in preparing shipboard marine pollution emergency plans **in conformity with** the **cited** regulations.

The main goal of these guidelines is to assist governments in **enacting** domestic laws which **give force to** and **implement** the cited regulations.

In the interest of uniformity, governments are requested **to refer to** these guidelines when preparing **appropriate** national regulations.